Name : Ragini Singh

Cohort : Jensen Huang

Roll No : 150096724023

Subject : System Design Final Exam Project

Sem : 4 , **Sprint :** 02

Batch : 2024 -2028

Problem Statement : Designing a Photo Sharing App like Instagram

Github Repo Link :

https://github.com/RaginiSingh2024/SnapShare-Photo-Sharing-Application

Deployment App Link

https://snapshare-photo-sharing-application-98lhmr5fzvwqsicdcedjuy.streamlit.app/

Documentation Link :

https://docs.google.com/document/d/1SfZv1fz5rc_DjhuVs2ElVloONnMHDBQg74Tt_u2oBpM/edit?usp=sharing

Python Source Code :

https://colab.research.google.com/drive/134sAPLI89m0VhbktT0ibSaj8cZyCB2bh?usp=sharing

SnapShare – Premium Photo Sharing Application

SnapShare

Modern Photo Sharing Platform Built with Streamlit, Python & SQLite

Instagram-inspired social media application with authentication, image uploads, social interactions, notifications, profile management, and theme customization.

Project Overview

SnapShare is a fully functional social media platform inspired by modern photo-sharing applications such as Instagram. The project demonstrates end-to-end system design principles including frontend development, backend processing, database management, media handling, and scalable architecture planning.

The application enables users to create accounts, upload images, interact with posts, manage profiles, receive notifications, and explore content through an intuitive user interface.

This project was developed as part of the **System Design Final Examination** and showcases practical implementation of software engineering, database design, and application architecture concepts.

Problem Statement

Traditional social media platforms require a robust architecture capable of handling user authentication, media uploads, content feeds, social interactions, notifications, and profile management.

The objective of SnapShare is to design and implement a scalable photo-sharing application that provides:

- Secure user authentication
- Media upload and optimization
- Personalized social feed
- User profile management
- Social interactions (likes, comments, follows)
- Notification tracking
- Responsive and modern user experience

Proposed Solution

SnapShare provides a layered architecture consisting of:

- Presentation Layer (Streamlit Frontend)
- Application Service Layer
- Database Layer (SQLite)
- File Storage Layer

The architecture separates concerns into independent modules, making the system maintainable, scalable, and easy to extend.

Key Features

Authentication System

- User Registration
- Secure Login
- Password Hashing
- Session Management
- Email Validation

Media Management

- Image Upload Support
- Automatic Compression
- Resolution Optimization
- JPEG Conversion
- Local Media Storage

Feed System

- Discover Feed
- Personalized Feed
- Chronological Post Ordering
- Trending Content Support

Social Interactions

- Like Posts

- Unlike Posts
- Comment System
- Follow Users
- Unfollow Users

Profile Management

- User Profiles
- Edit Profile
- Bio Management
- Profile Statistics
- Profile Picture Updates

Notification System

- Like Notifications
- Comment Notifications
- Follow Notifications
- Unread Notification Tracking

UI Features

- Dark Theme
- Light Theme
- Responsive Layout
- Custom CSS Styling
- Modern Glassmorphism Design

🏗️ System Architecture

High-Level Architecture

```
USER
  |
  ▼
STREAMLIT FRONTEND
  |
  ▼
APPLICATION LAYER
  |
  ├── Authentication Module
  ├── Feed Management Module
  ├── Post Management Module
  ├── Social Interaction Module
  ├── Profile Management Module
  ├── Notification Module
  └── Theme Engine
   |
   ▼
DATABASE LAYER (SQLite)
  |
  ▼
FILE STORAGE
```

Architecture Diagram

Replace with your uploaded architecture image:

`![System Architecture](SnapShare_Final_Architecture.png)`

Database Design

Users Table

- user_id
- username
- email
- password
- bio
- profile_picture
- created_at

Posts Table

- post_id
- user_id
- image_path
- caption
- created_at

Likes Table

- like_id
- user_id
- post_id
- created_at

Comments Table

- comment_id

- user_id
- post_id
- comment
- created_at

Followers Table

- follower_id
- following_id
- created_at

Notifications Table

- notification_id
- user_id
- actor_id
- type
- post_id
- created_at

Technology Stack

Component	Technology
Frontend	Streamlit
Backend	Python
Database	SQLite

Storage	Local Upload Folder
Styling	Custom CSS
Image Processing	Pillow
Version Control	Git & GitHub

📂 Project Structure

```
SnapShare/
│
├── app.py
├── auth.py
├── database.py
├── feed.py
├── posts.py
```

```
├── profile.py
├── themes.py
│
├── uploads/
│
├── README.md
├── requirements.txt
├── snapshare.db
│
├── SnapShare_Final_Architecture.png
└── system_design_report.md
```

Installation Guide

Clone Repository

git clone
https://github.com/RaginiSingh2024/SnapShare-Photo-Sharing-Application.git

cd SnapShare-Photo-Sharing-Application

Install Dependencies

pip install -r requirements.txt

Run Application

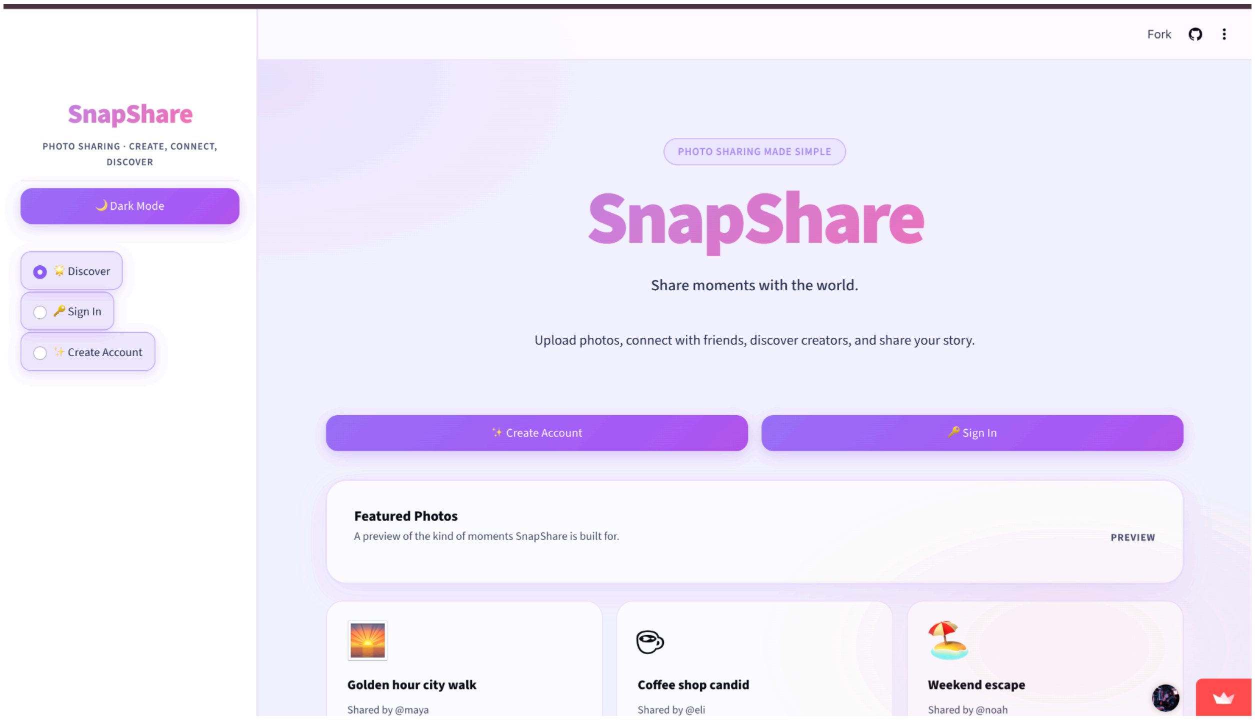
streamlit run app.py

Open:

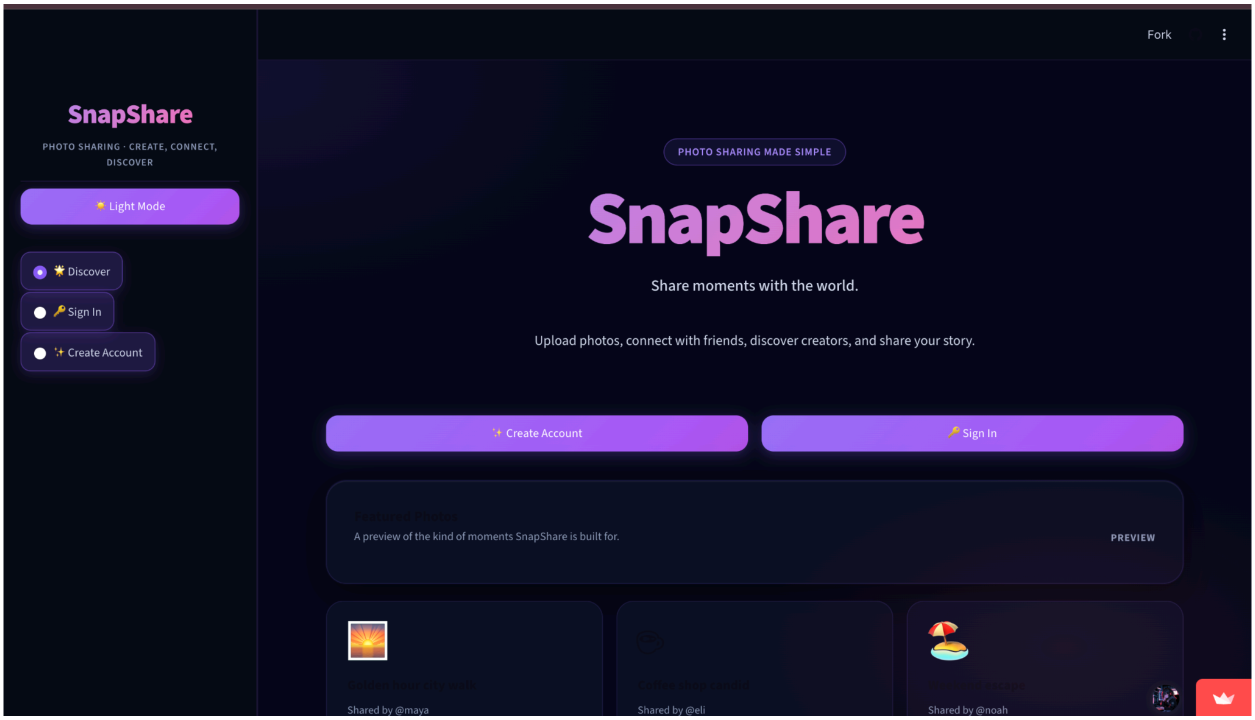
http://localhost:8501

Screenshots

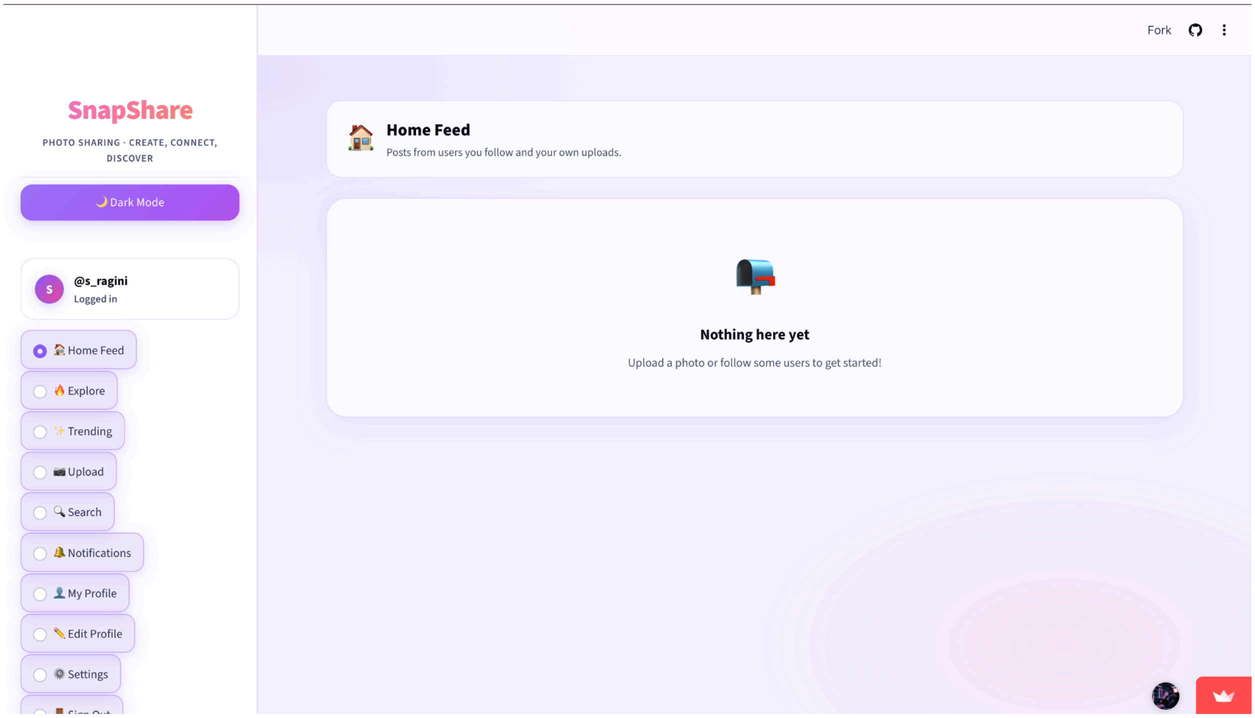
1. **Dashboard**



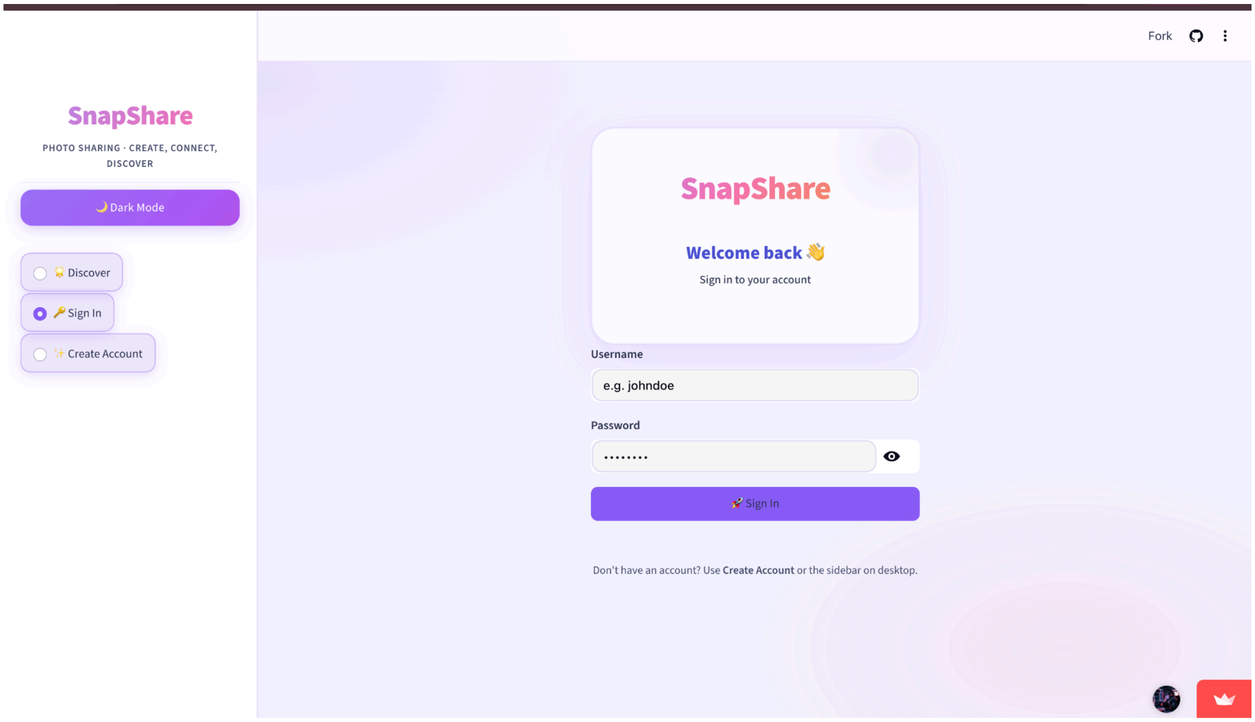
2. Dashboard Dark Mode


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Light Mode
Discover
Sign In
Create Account
PHOTO SHARING MADE SIMPLE
SnapShare
Share moments with the world.
Upload photos, connect with friends, discover creators, and share your story.
Create Account
Sign In
A preview of the kind of moments SnapShare is built for.
PREVIEW
Shared by @maya
Shared by @eli
Shared by @noah

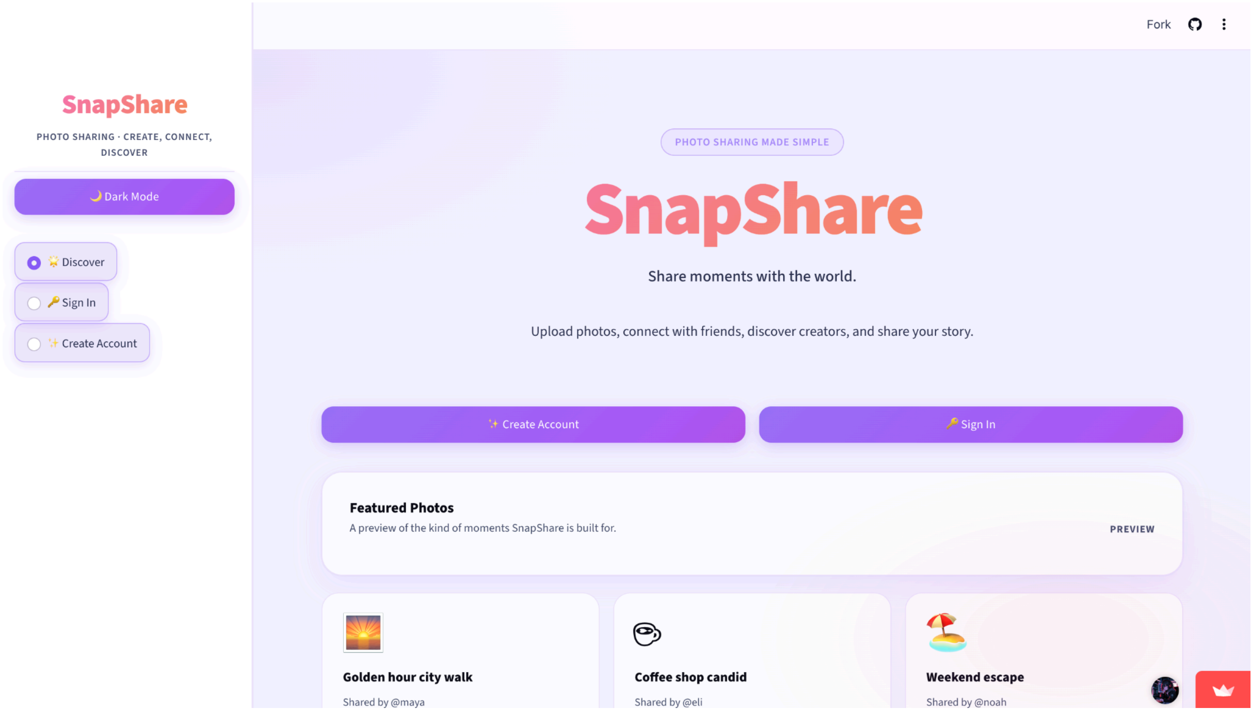
Home feed


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Home Feed
Posts from users you follow and your own uploads.
Nothing here yet
Upload a photo or follow some users to get started!

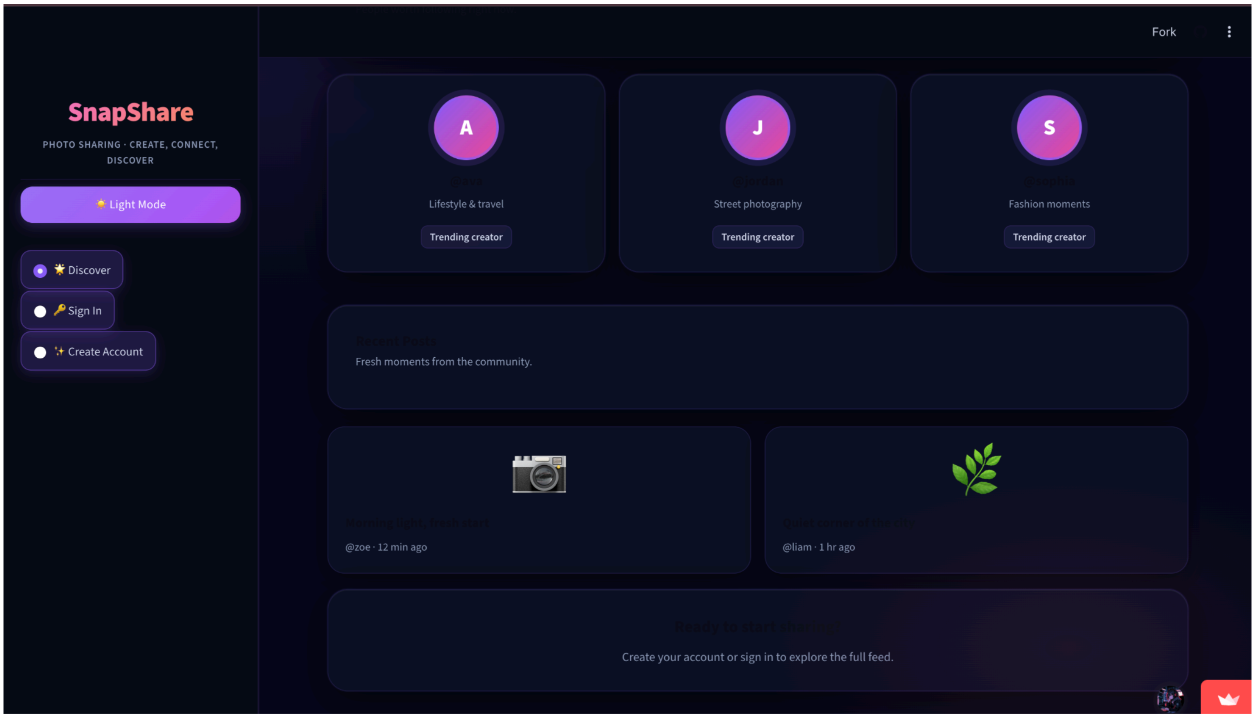
Sign-in page

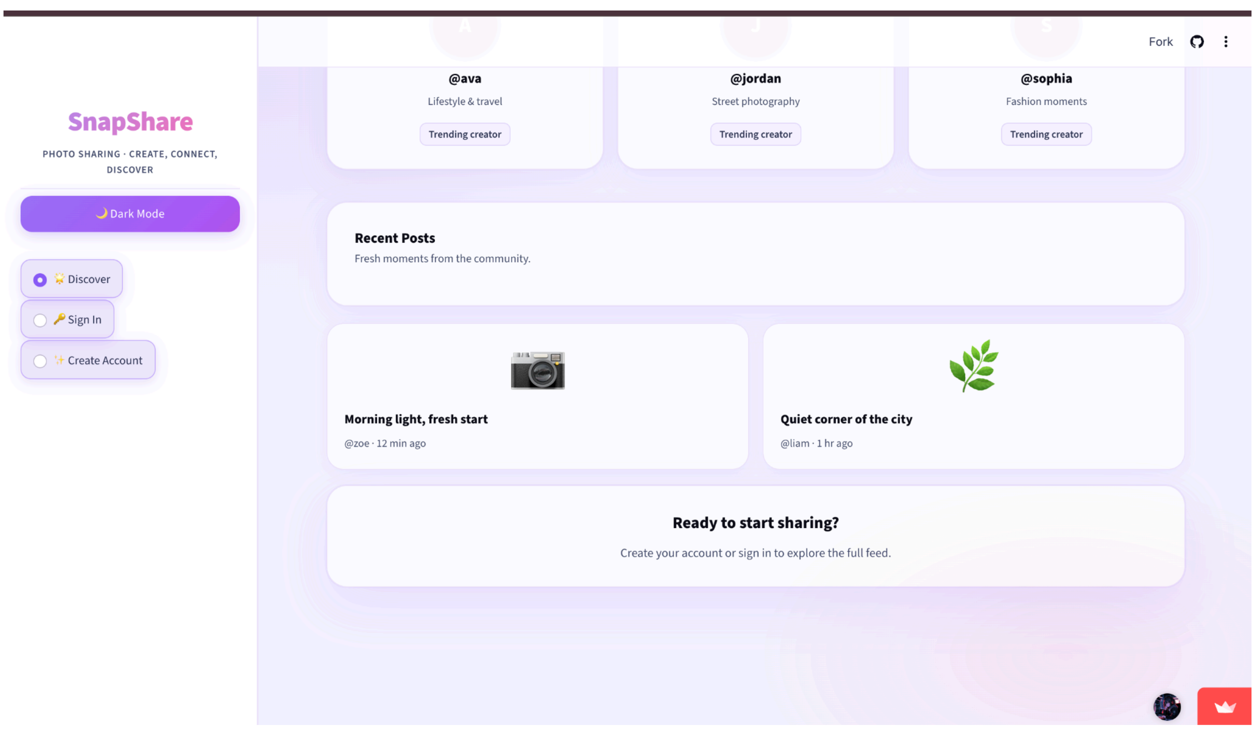


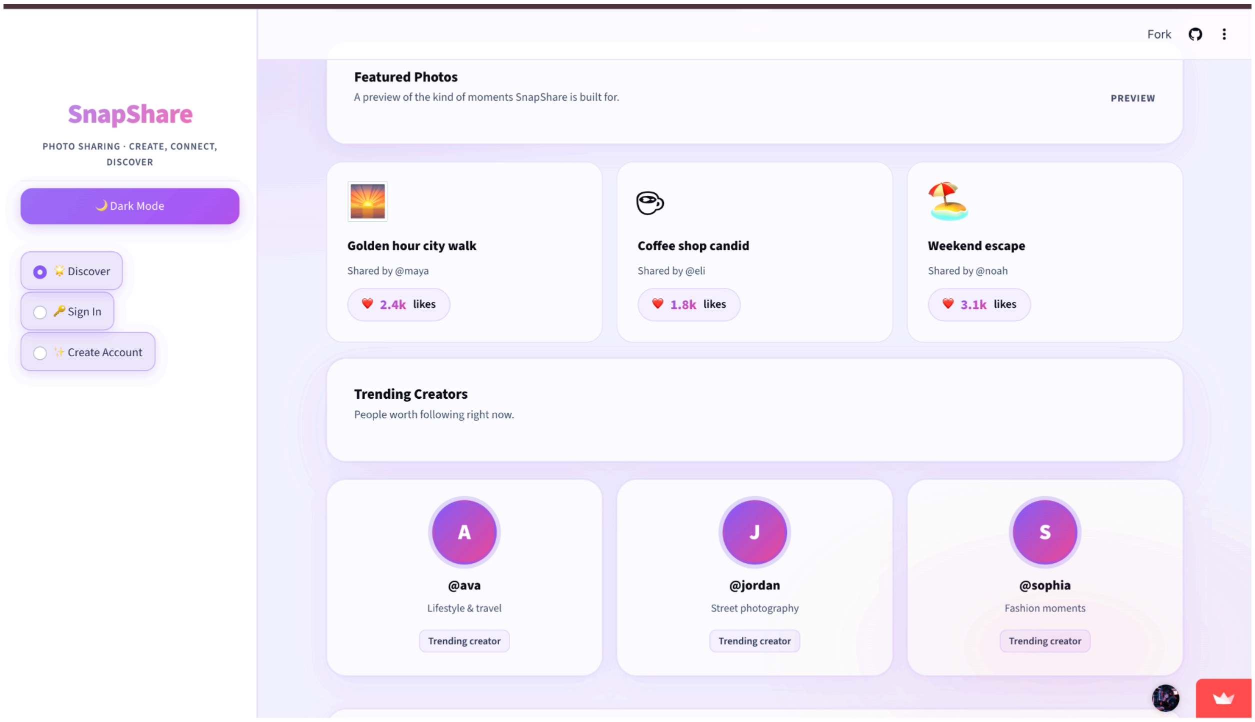
5 Discover page



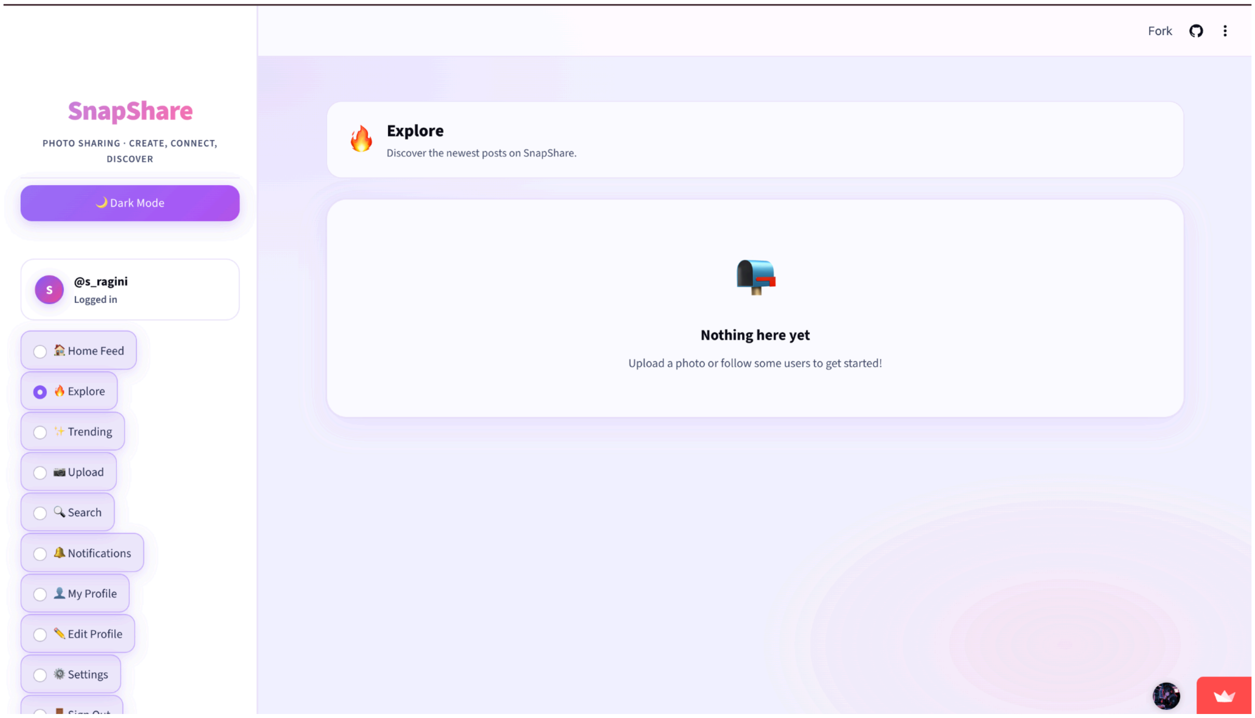
6. Discover Dark mode

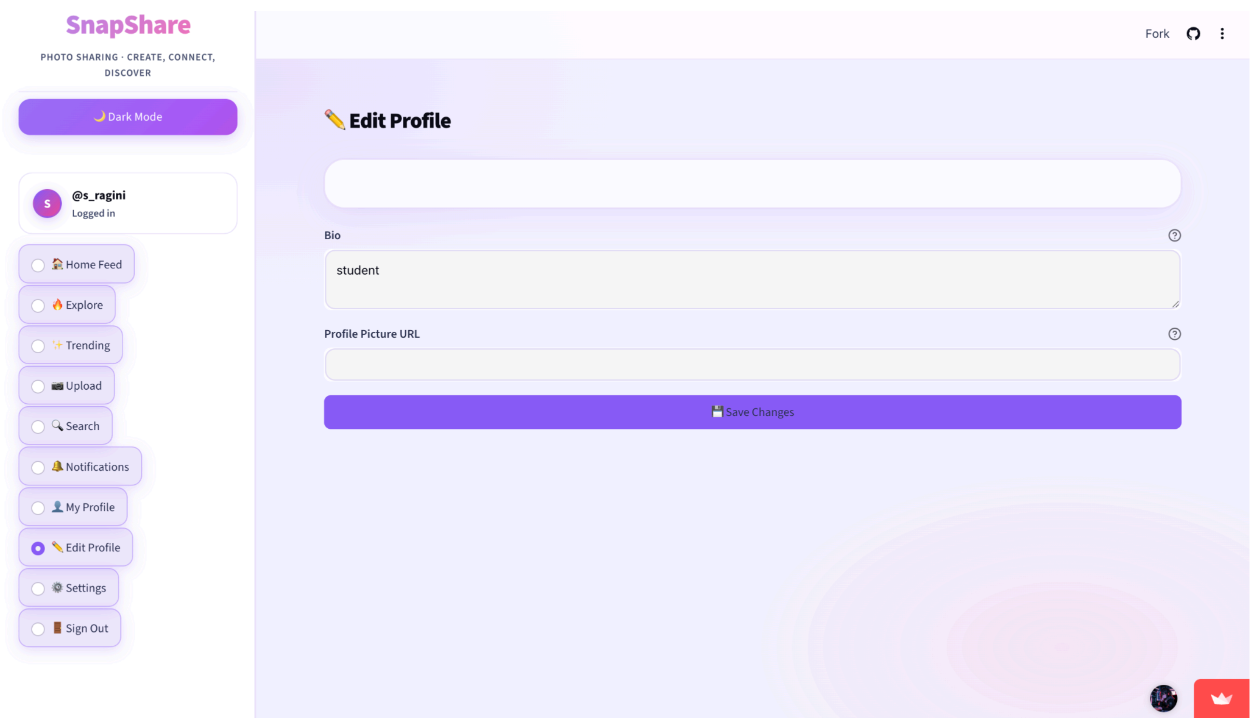

Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Light Mode
Discover
Sign In
Create Account
A
Lifestyle & travel
Trending creator
J
Street photography
Trending creator
S
Fashion moments
Trending creator
Fresh moments from the community.
@zoe · 12 min ago
@liam · 1 hr ago
Create your account or sign in to explore the full feed.


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
Discover
Sign In
Create Account
@ava
Lifestyle & travel
Trending creator
@jordan
Street photography
Trending creator
@sophia
Fashion moments
Trending creator
Recent Posts
Fresh moments from the community.
Morning light, fresh start
@zoe · 12 min ago
Quiet corner of the city
@liam · 1 hr ago
Ready to start sharing?
Create your account or sign in to explore the full feed.



Explore feed


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Explore
Discover the newest posts on SnapShare.
Nothing here yet
Upload a photo or follow some users to get started!


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Fork
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
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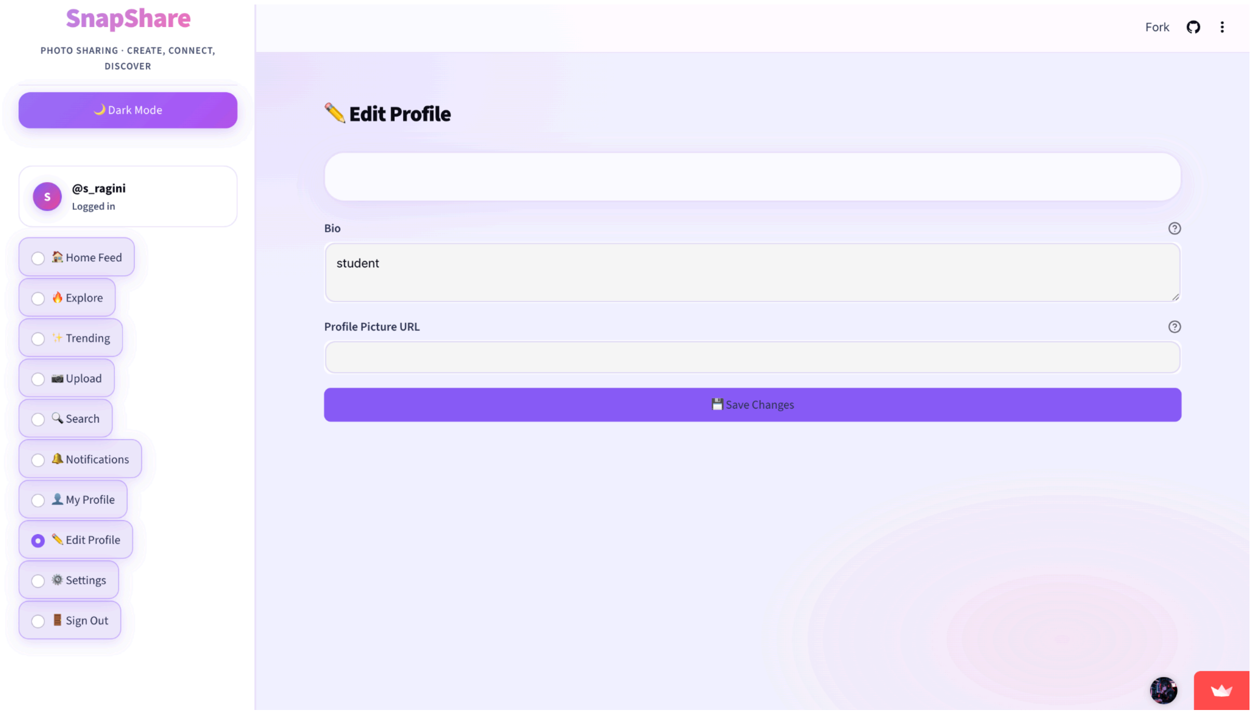
Notifications
My Profile
Edit Profile
Settings
Sign Out
Edit Profile
Bio
student
Profile Picture URL
Save Changes


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
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student
Profile Picture URL
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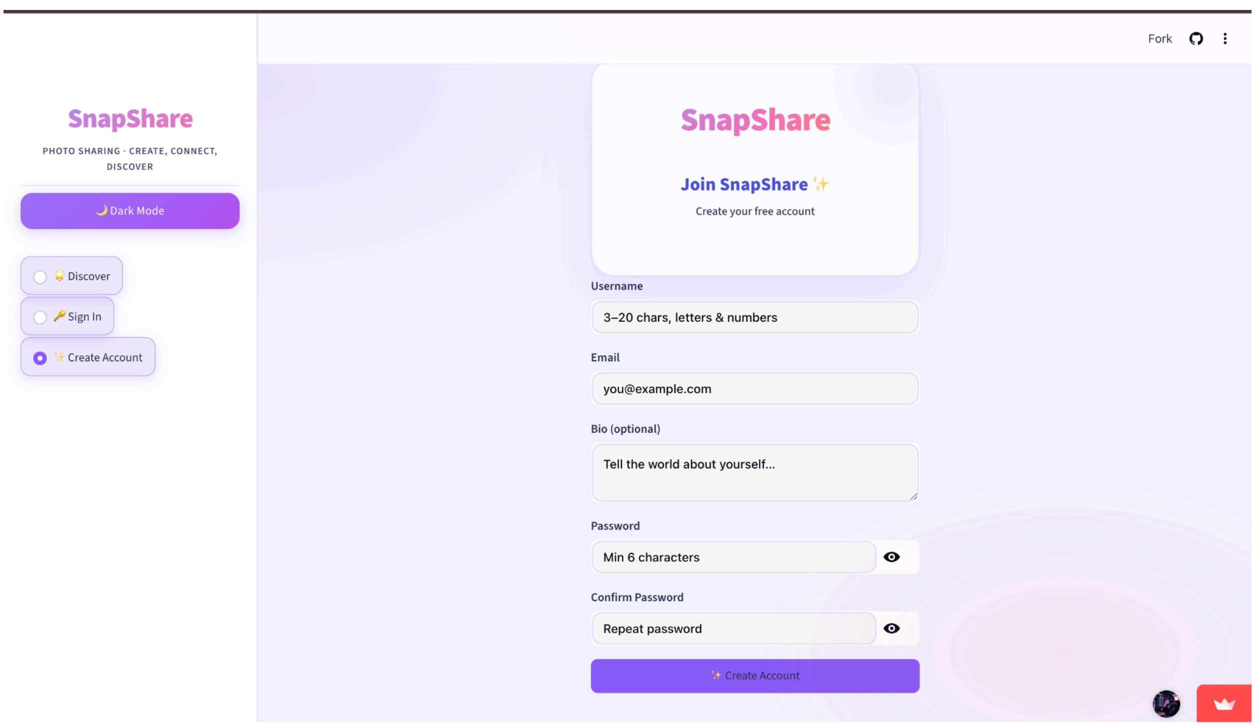
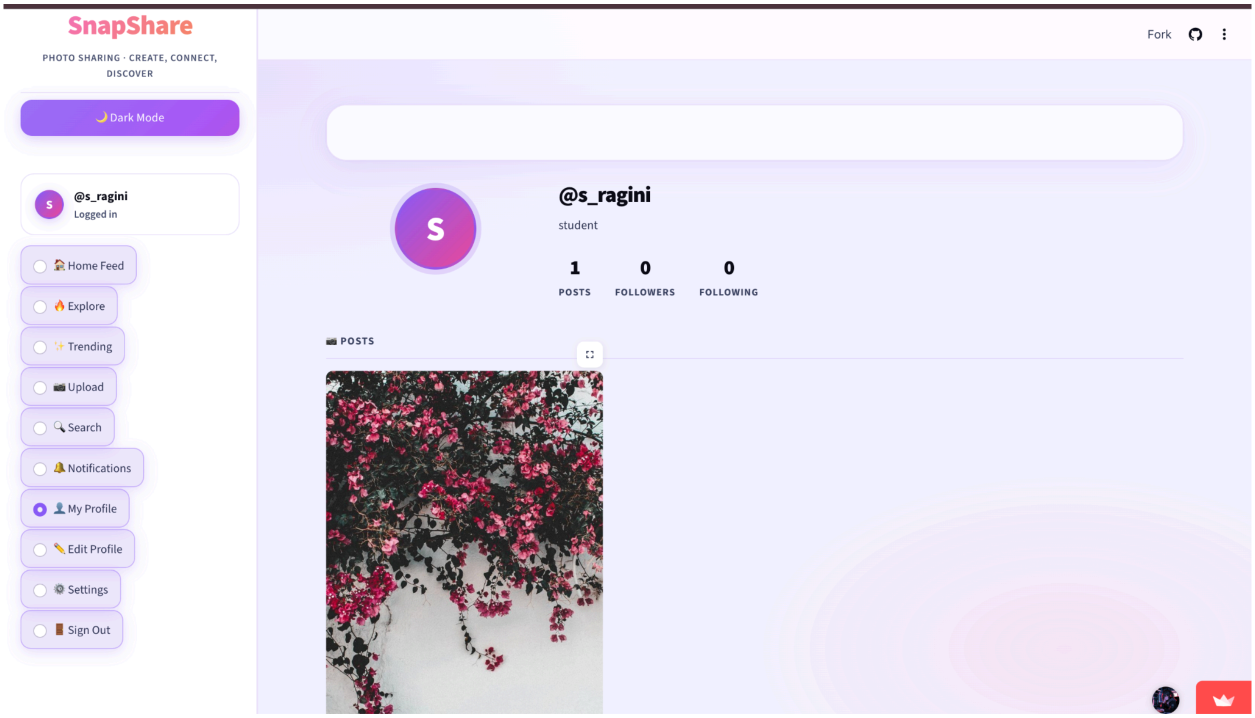

SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
Discover
Sign In
Create Account
Fork
SnapShare
Join SnapShare
Create your free account
Username
3–20 chars, letters & numbers
Email
you@example.com
Bio (optional)
Tell the world about yourself...
Password
Min 6 characters
Confirm Password
Repeat password
Create Account


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
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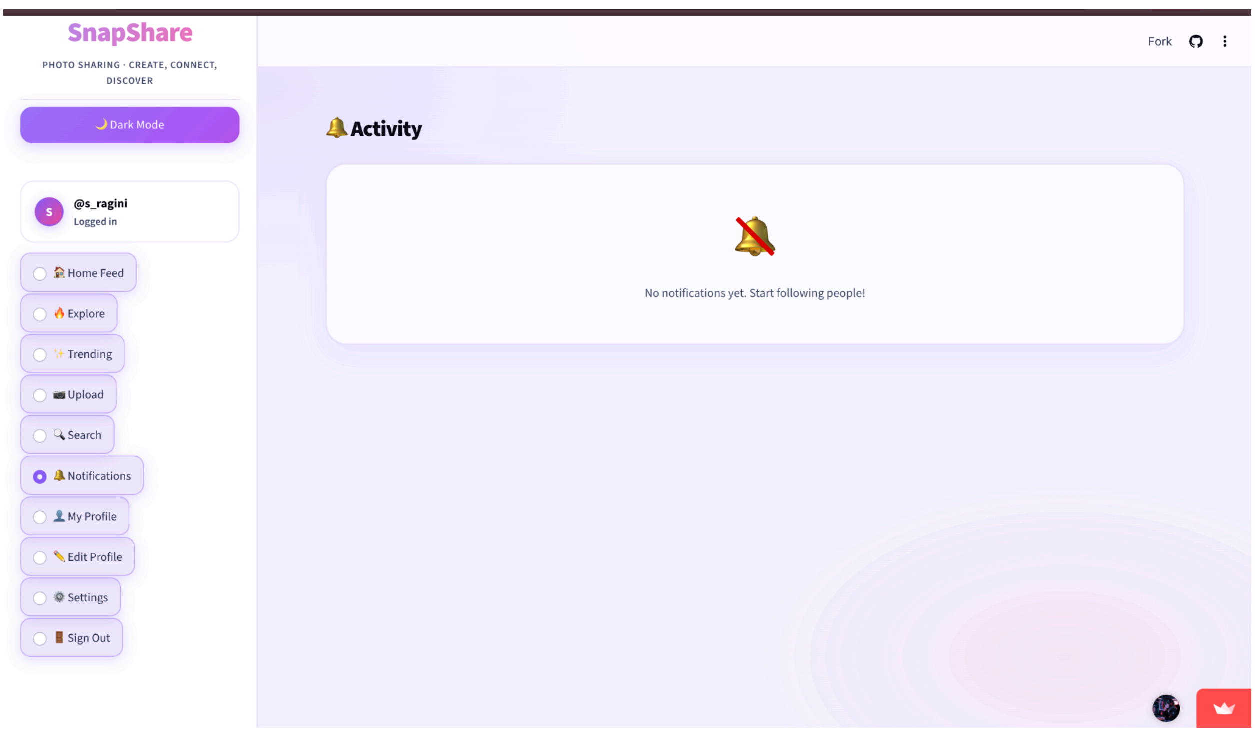
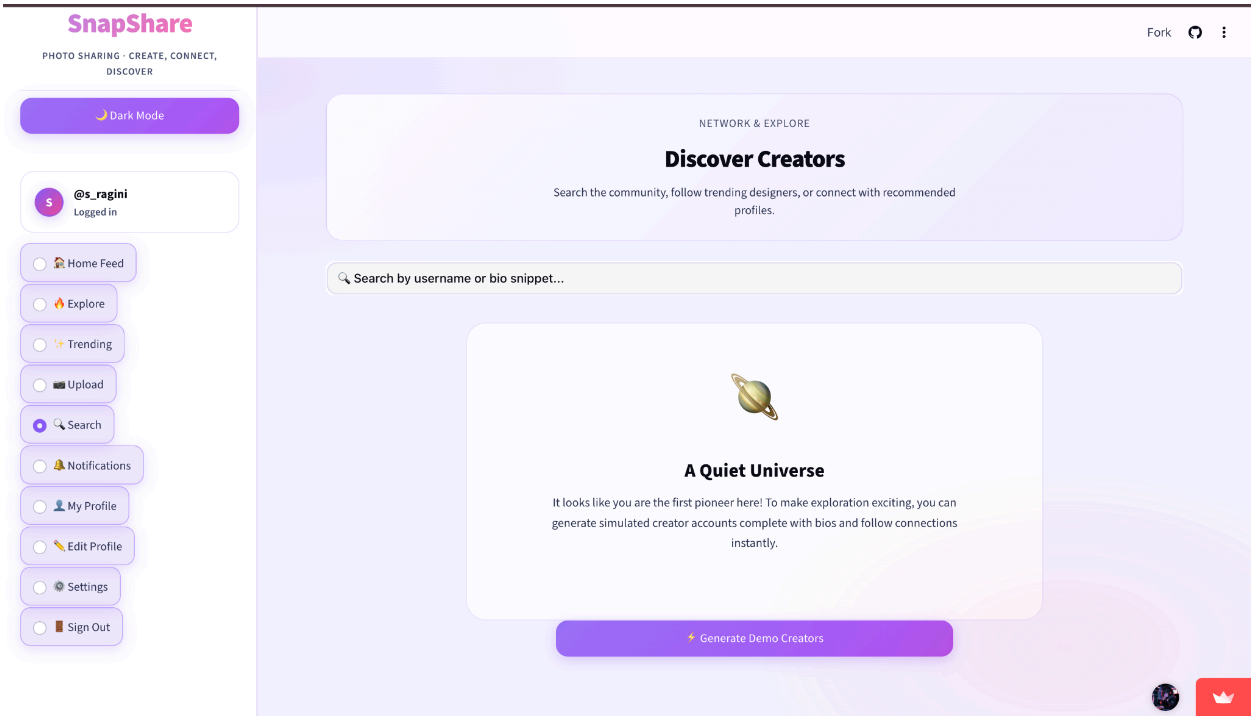
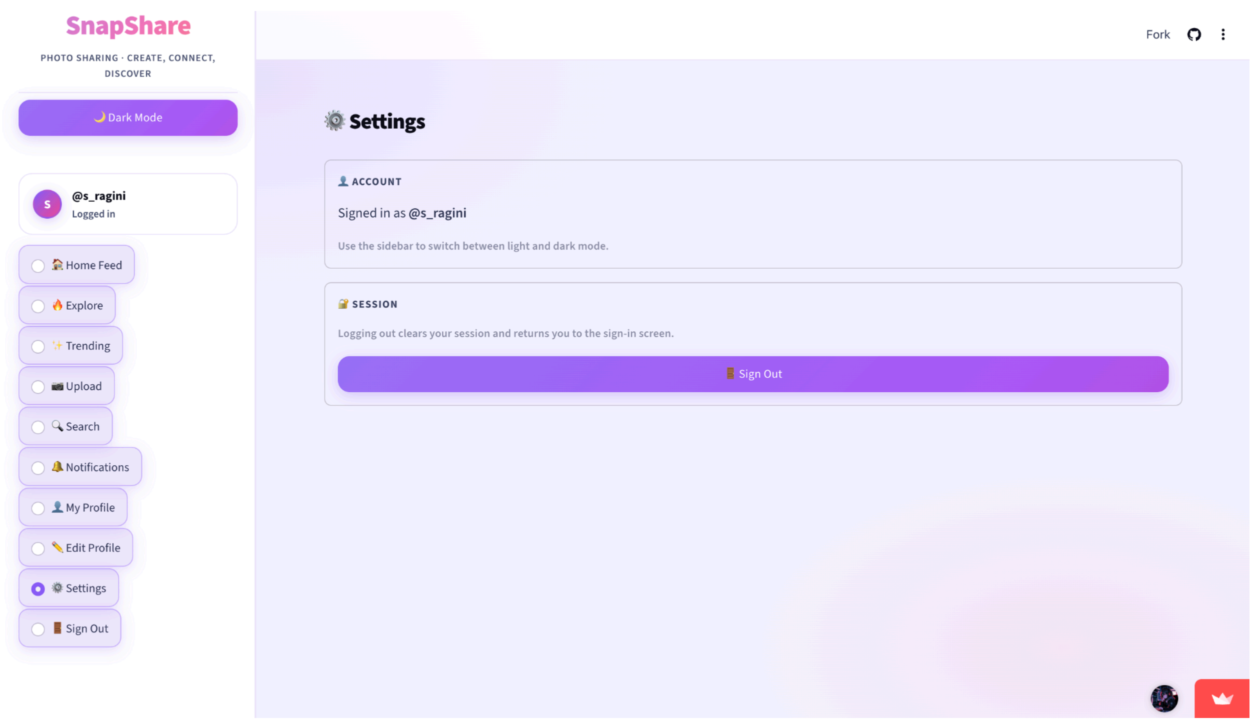
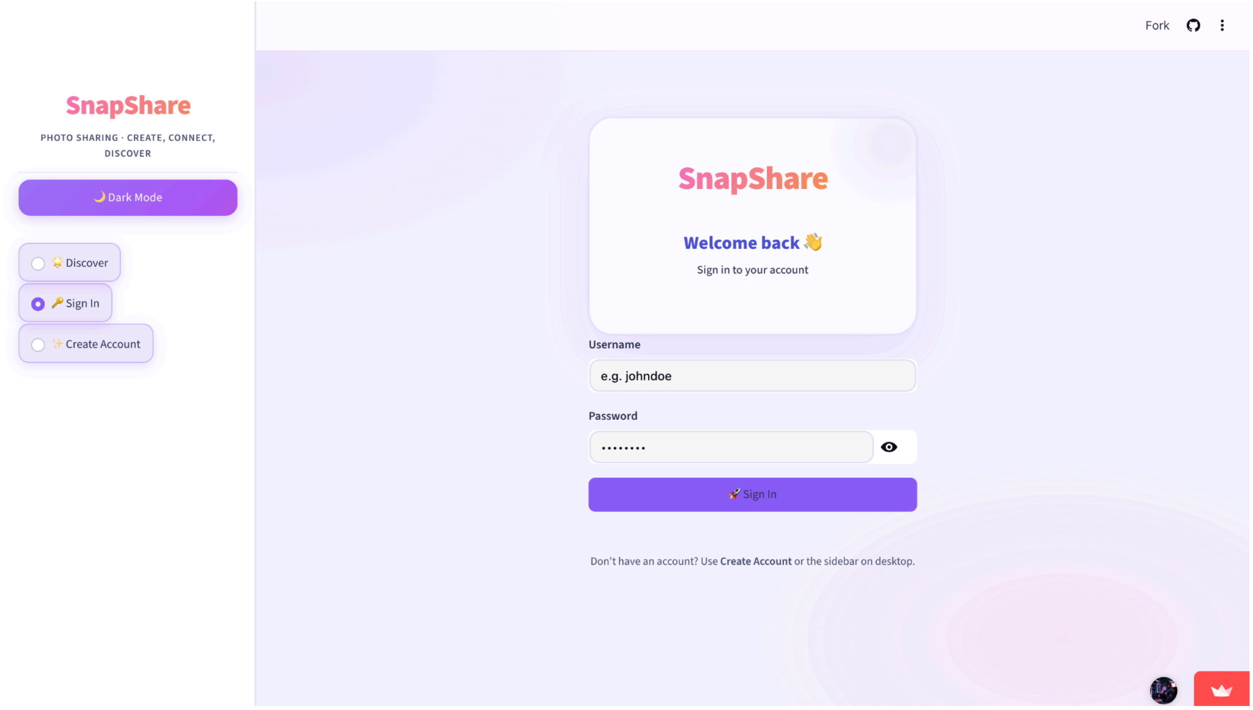
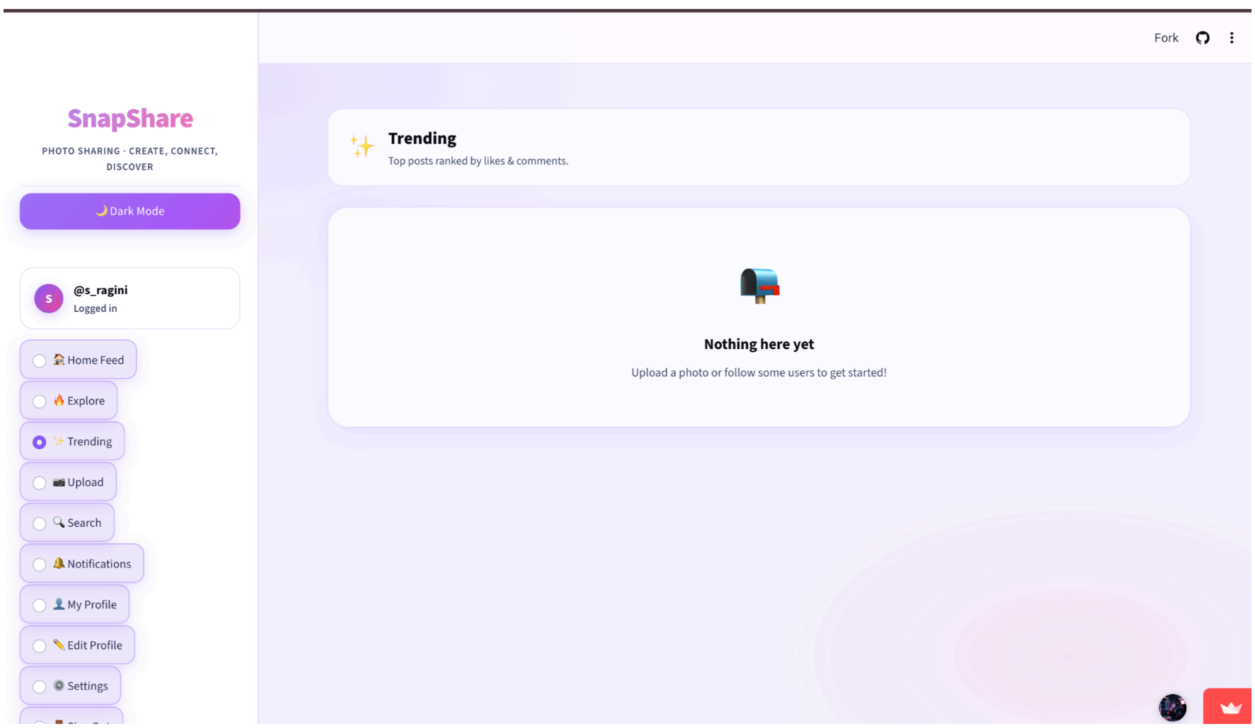
Notifications
My Profile
Edit Profile
Settings
Sign Out
Fork
@s_ragini
student
1
POSTS
0
FOLLOWERS
0
FOLLOWING
POSTS


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Sign Out
Fork
Activity
No notifications yet. Start following people!


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Sign Out
Fork
NETWORK & EXPLORE
Discover Creators
Search the community, follow trending designers, or connect with recommended profiles.
Search by username or bio snippet...
A Quiet Universe
It looks like you are the first pioneer here! To make exploration exciting, you can generate simulated creator accounts complete with bios and follow connections instantly.
Generate Demo Creators


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Sign Out
Fork
Settings
ACCOUNT
Signed in as @s_ragini
Use the sidebar to switch between light and dark mode.
SESSION
Logging out clears your session and returns you to the sign-in screen.
Sign Out


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
Discover
Sign In
Create Account
SnapShare
Welcome back 👋
Sign in to your account
Username
e.g. johndoe
Password
Sign In
Don't have an account? Use Create Account or the sidebar on desktop.


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
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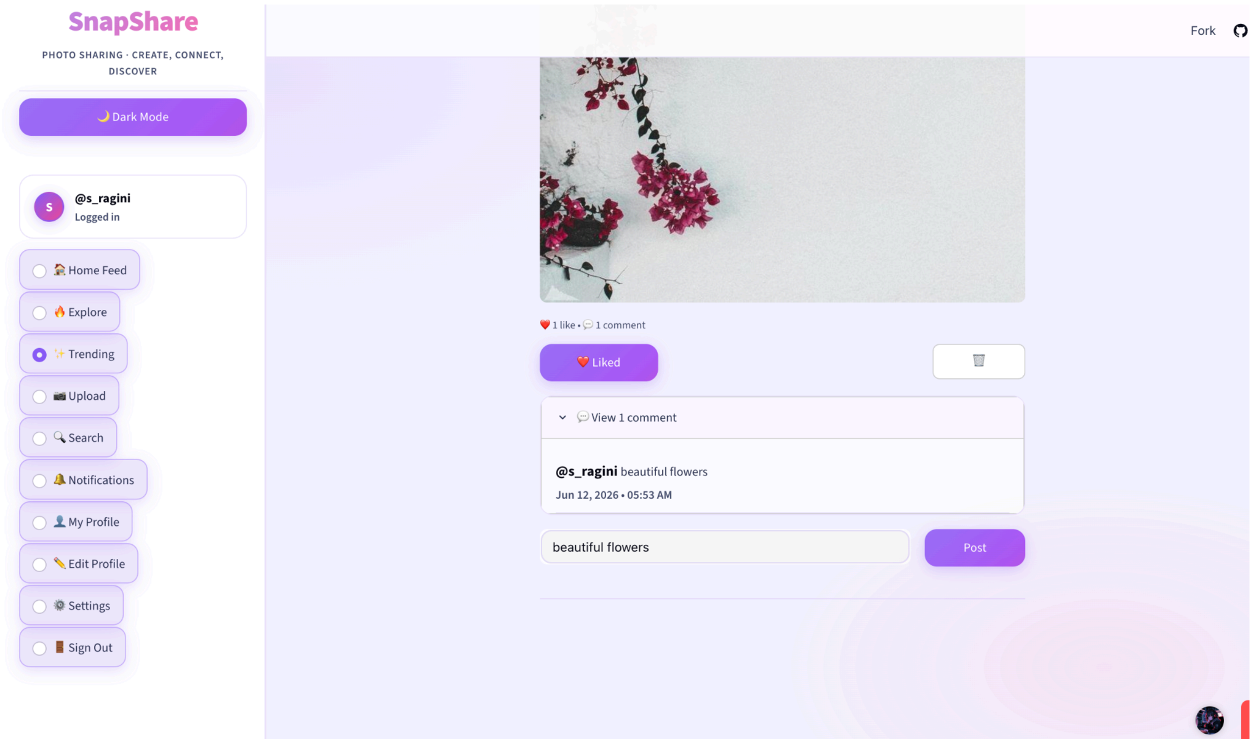
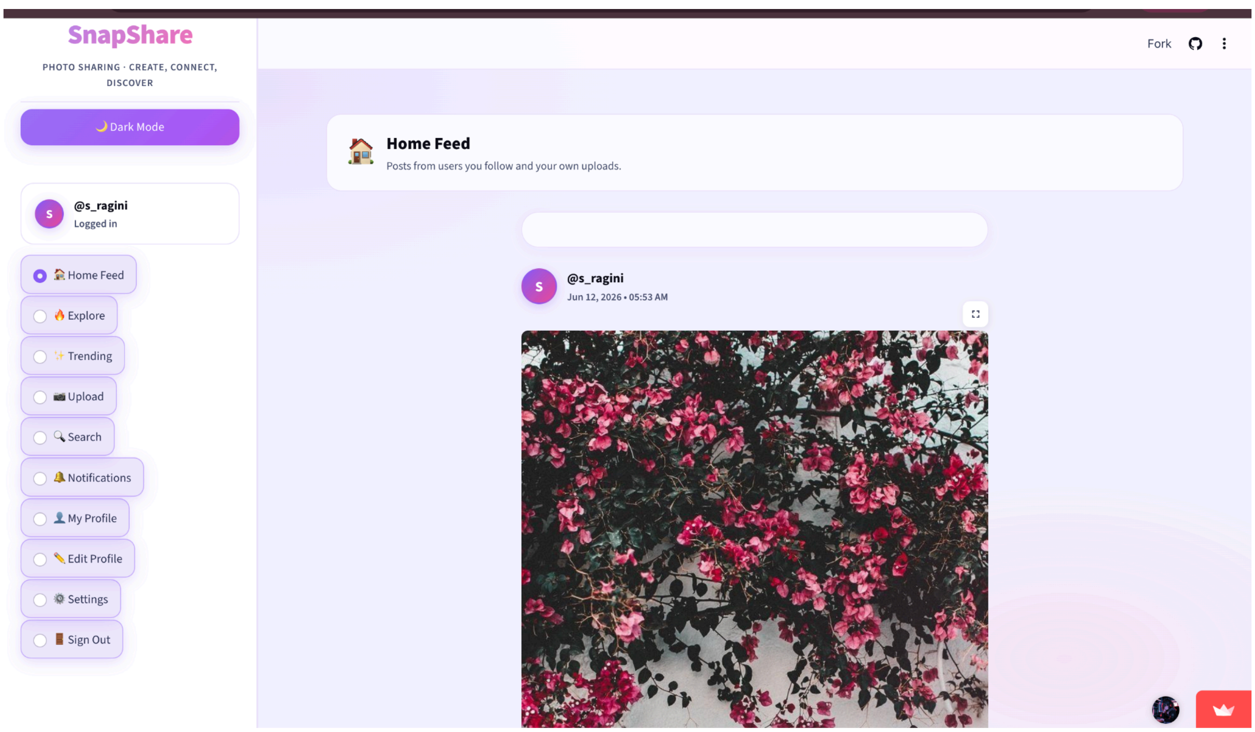
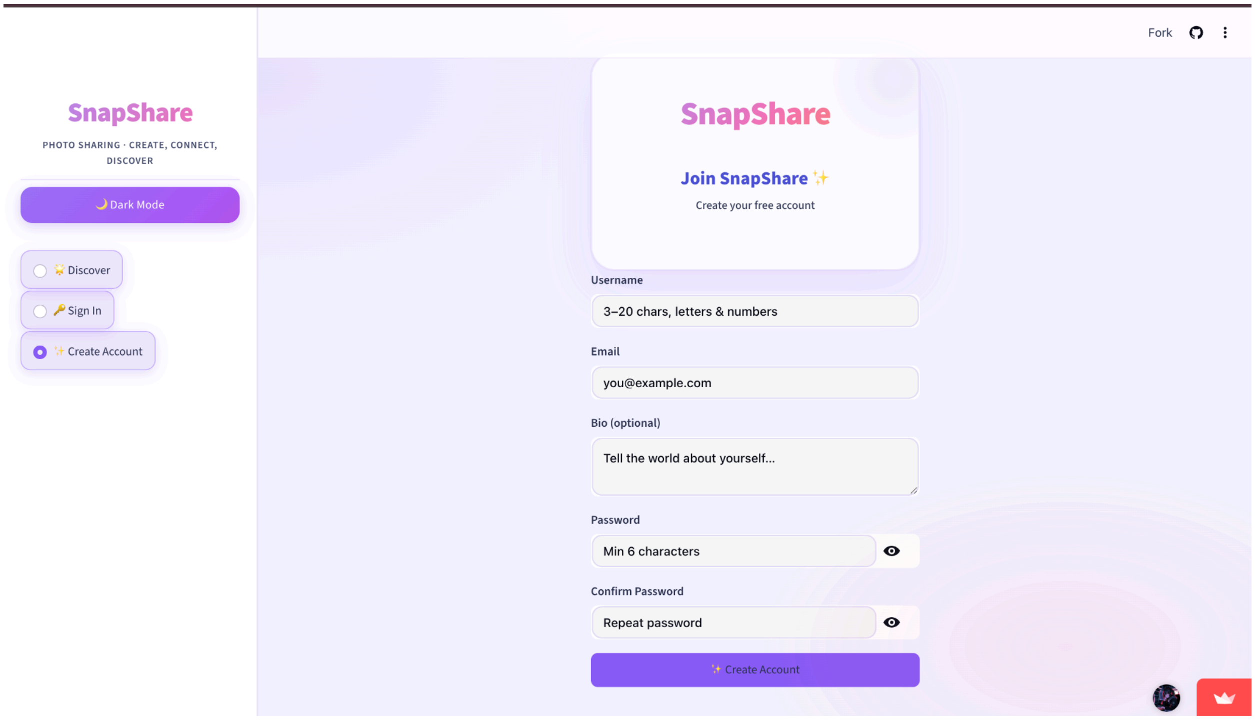
S
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Trending
Top posts ranked by likes & comments.
Nothing here yet
Upload a photo or follow some users to get started!

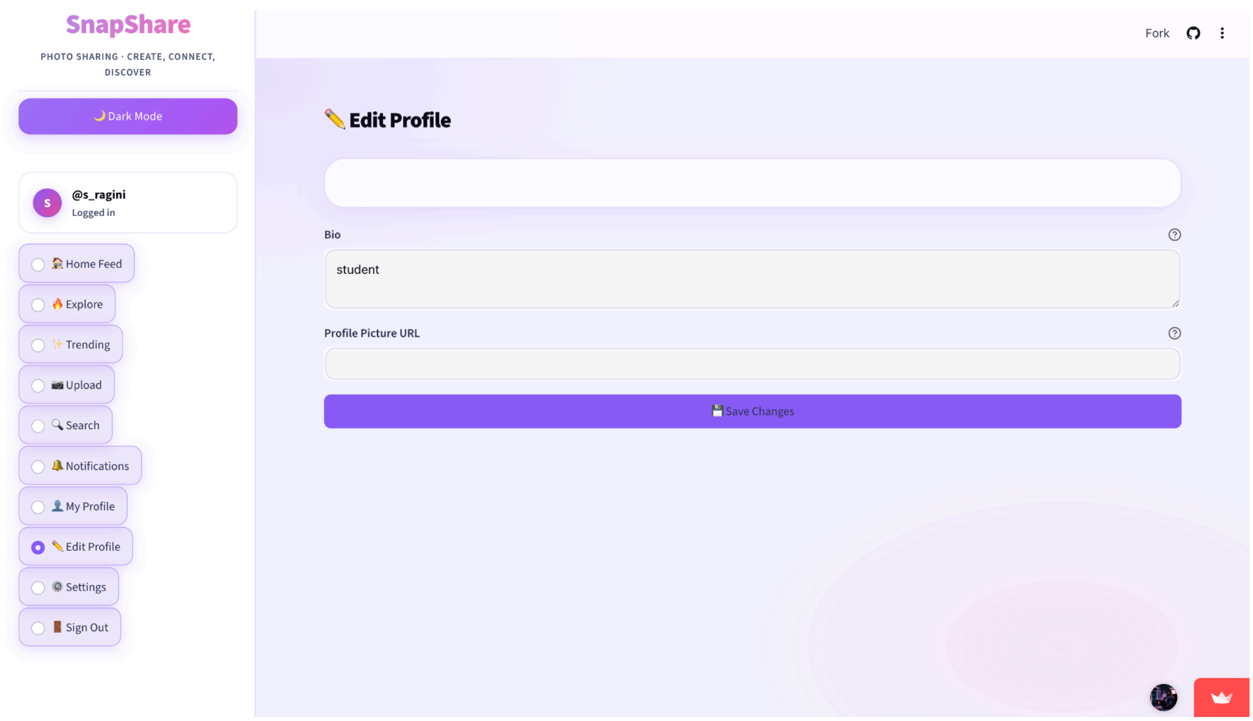

SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
@s_ragini
Logged in
🏠 Home Feed
🔥 Explore
✨ Trending
📷 Upload
🔍 Search
🔔 Notifications
👤 My Profile
✏️ Edit Profile
⚙️ Settings
🚪 Sign Out
Fork
❤️ 1 like • 💬 1 comment
❤️ Liked
💬 View 1 comment
@s_ragini beautiful flowers
Jun 12, 2026 • 05:53 AM
beautiful flowers
Post


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
@s_ragini
Logged in
🏠 Home Feed
🔥 Explore
✨ Trending
📷 Upload
🔍 Search
🔔 Notifications
👤 My Profile
✏️ Edit Profile
⚙️ Settings
🚪 Sign Out
Fork
Home Feed
Posts from users you follow and your own uploads.
@s_ragini
Jun 12, 2026 • 05:53 AM

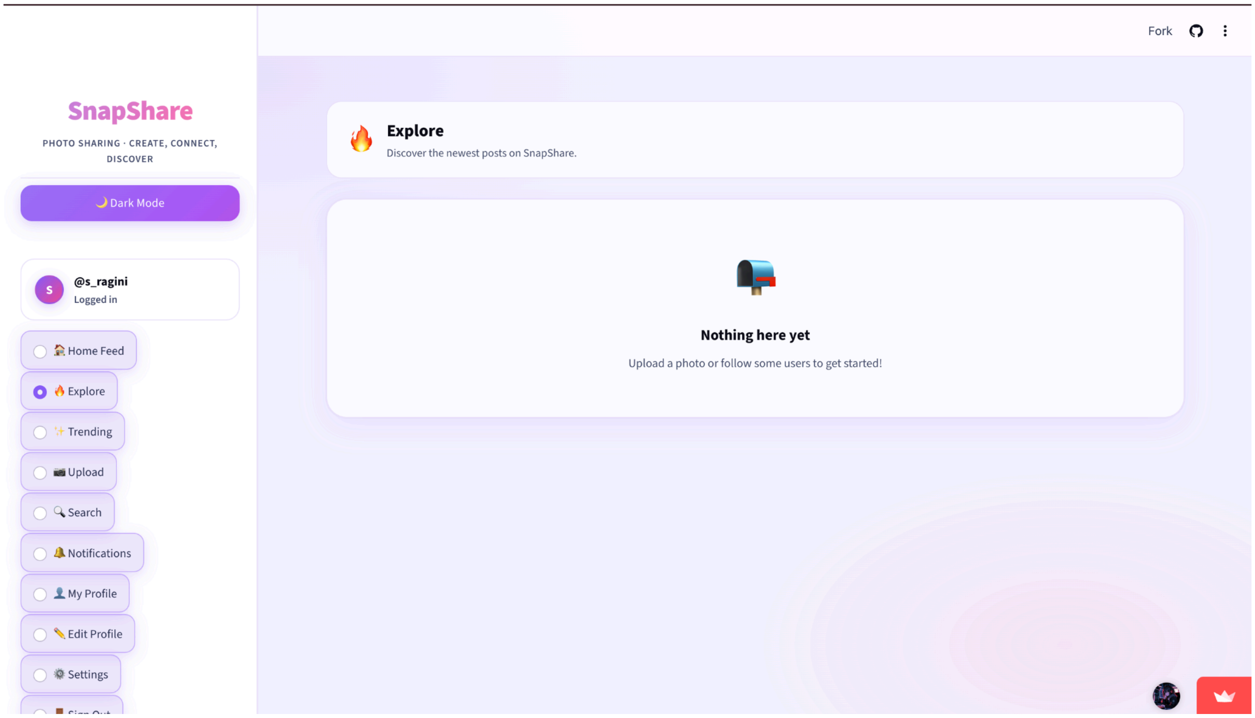
Account png

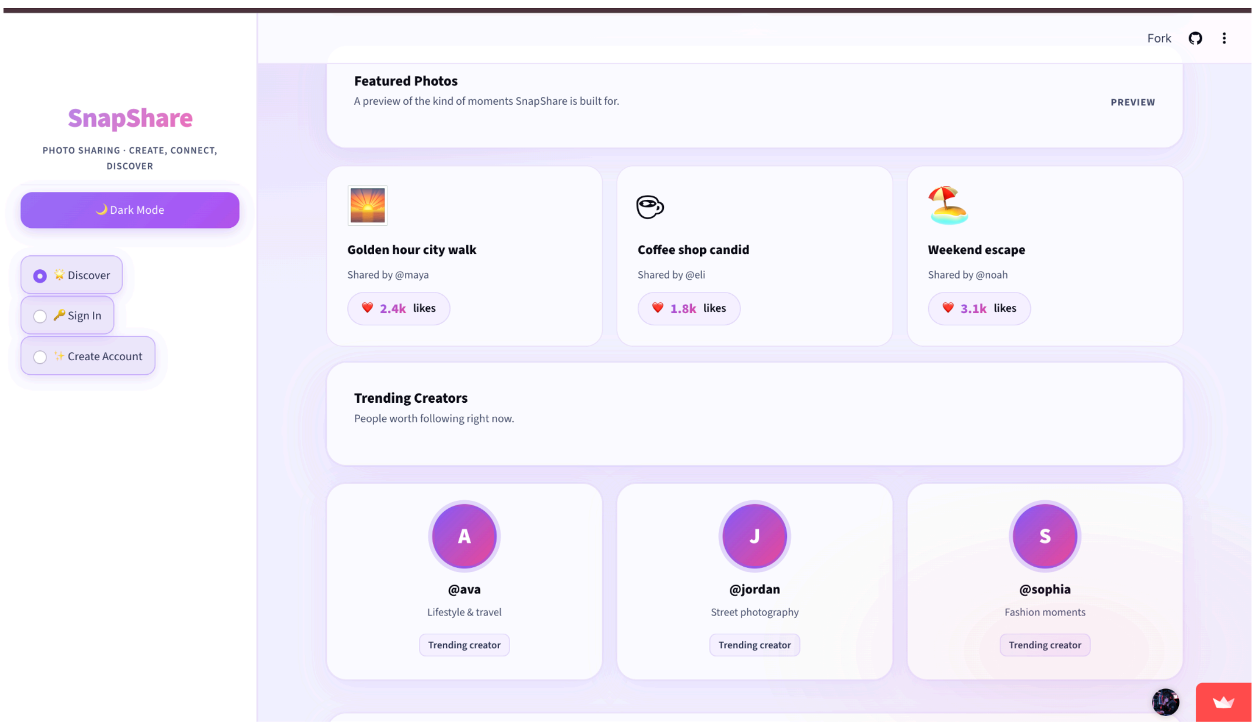


Edit profile

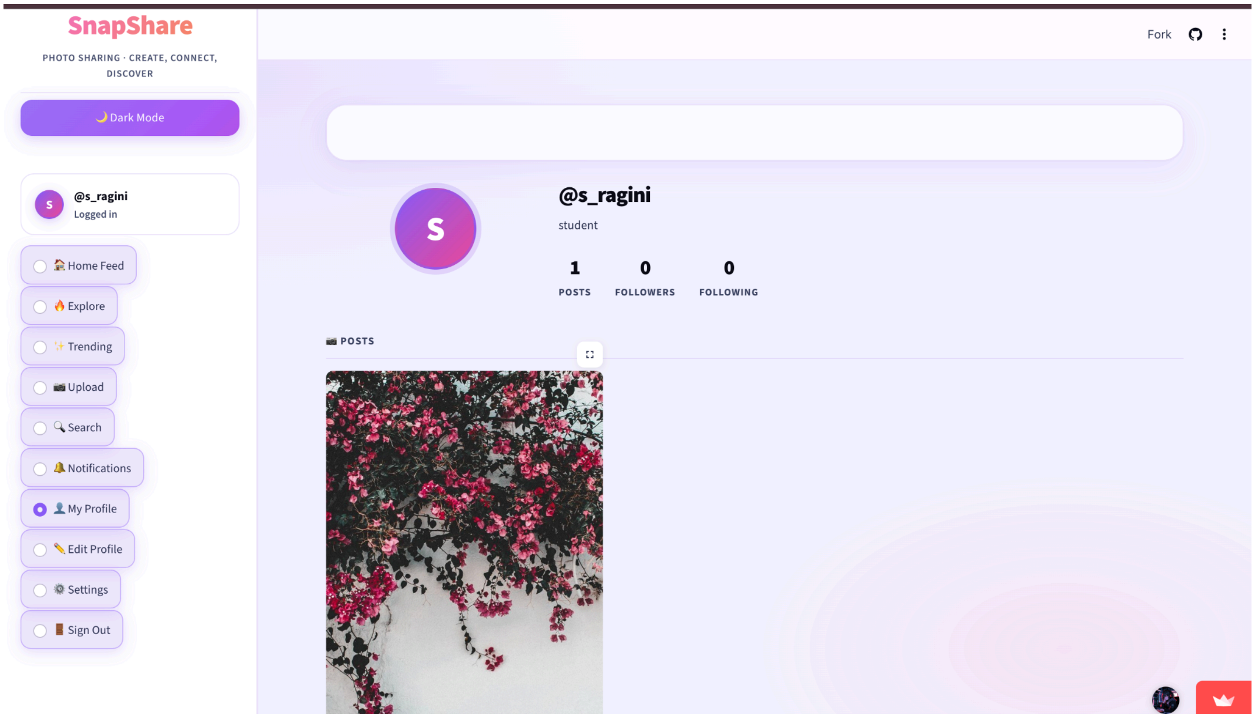


Explore png

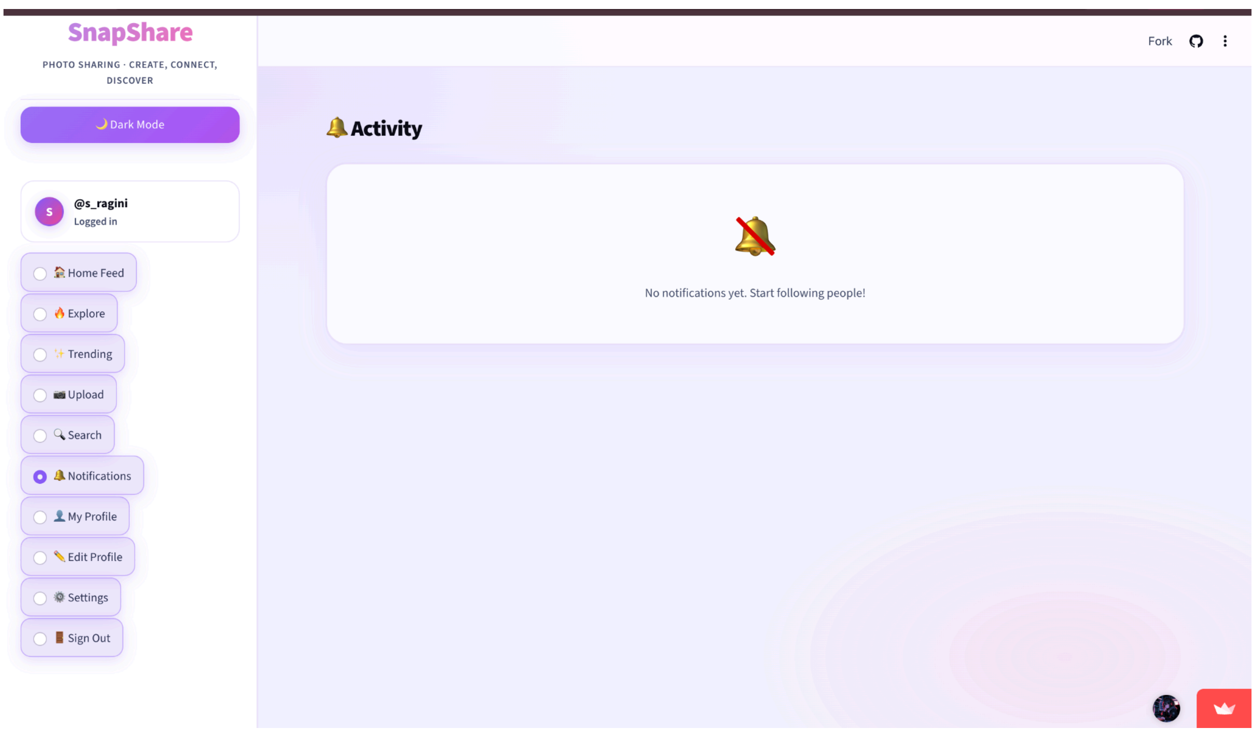

Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
@s_ragini
Logged in
Home Feed
Explore
Trending
Upload
Search
Notifications
My Profile
Edit Profile
Settings
Explore
Discover the newest posts on SnapShare.
Nothing here yet
Upload a photo or follow some users to get started!


Fork
SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
Dark Mode
Discover
Sign In
Create Account
Featured Photos
A preview of the kind of moments SnapShare is built for.
PREVIEW
Golden hour city walk
Shared by @maya
2.4k likes
Coffee shop candid
Shared by @eli
1.8k likes
Weekend escape
Shared by @noah
3.1k likes
Trending Creators
People worth following right now.
A
@ava
Lifestyle & travel
Trending creator
J
@jordan
Street photography
Trending creator
S
@sophia
Fashion moments
Trending creator

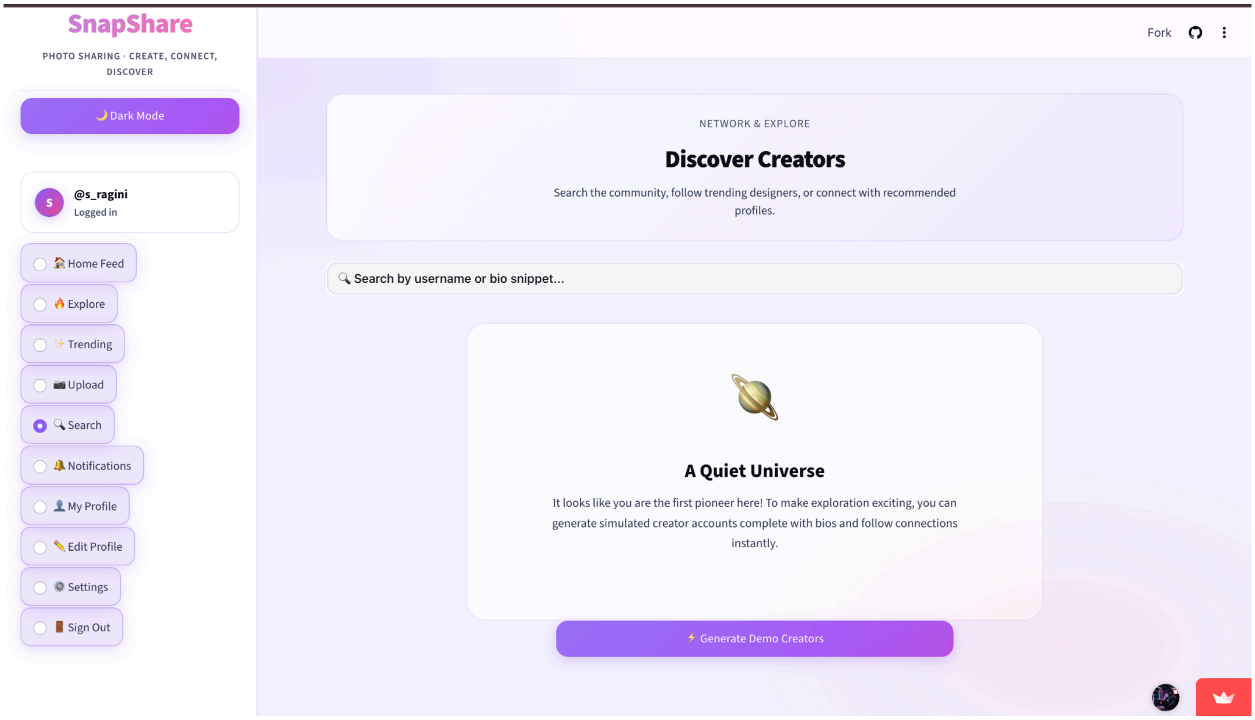
My profile feed



Notification

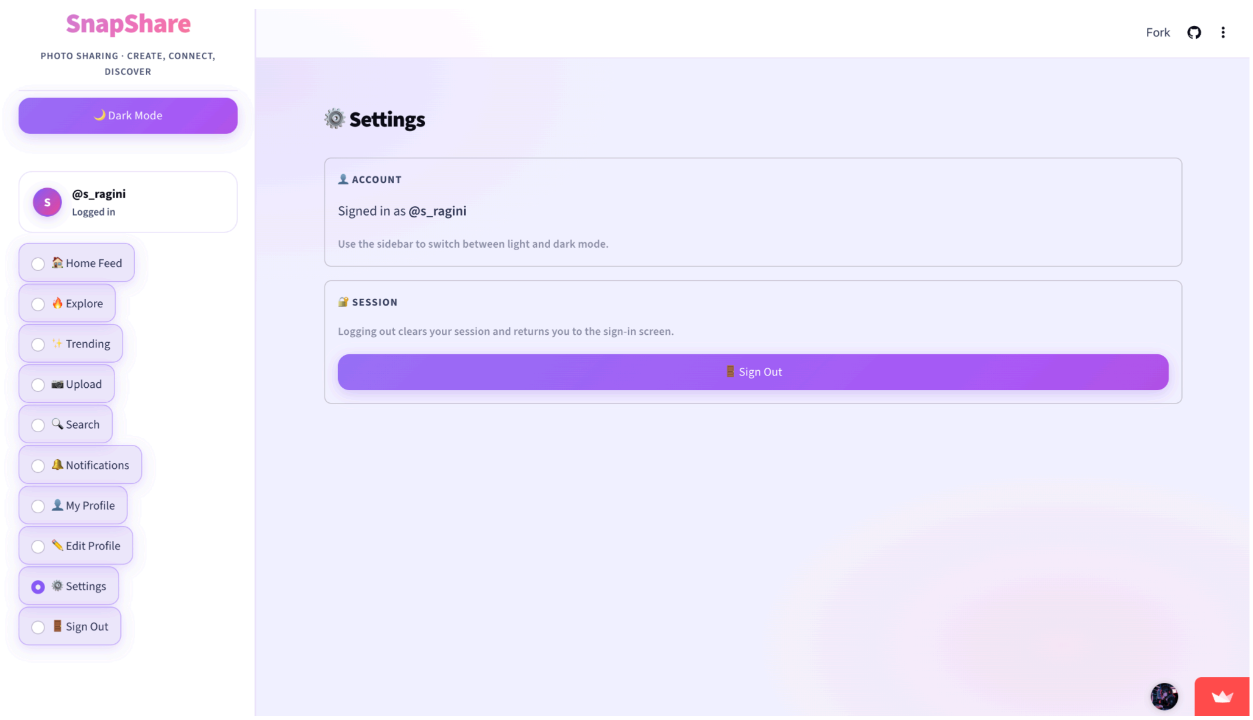


Search page

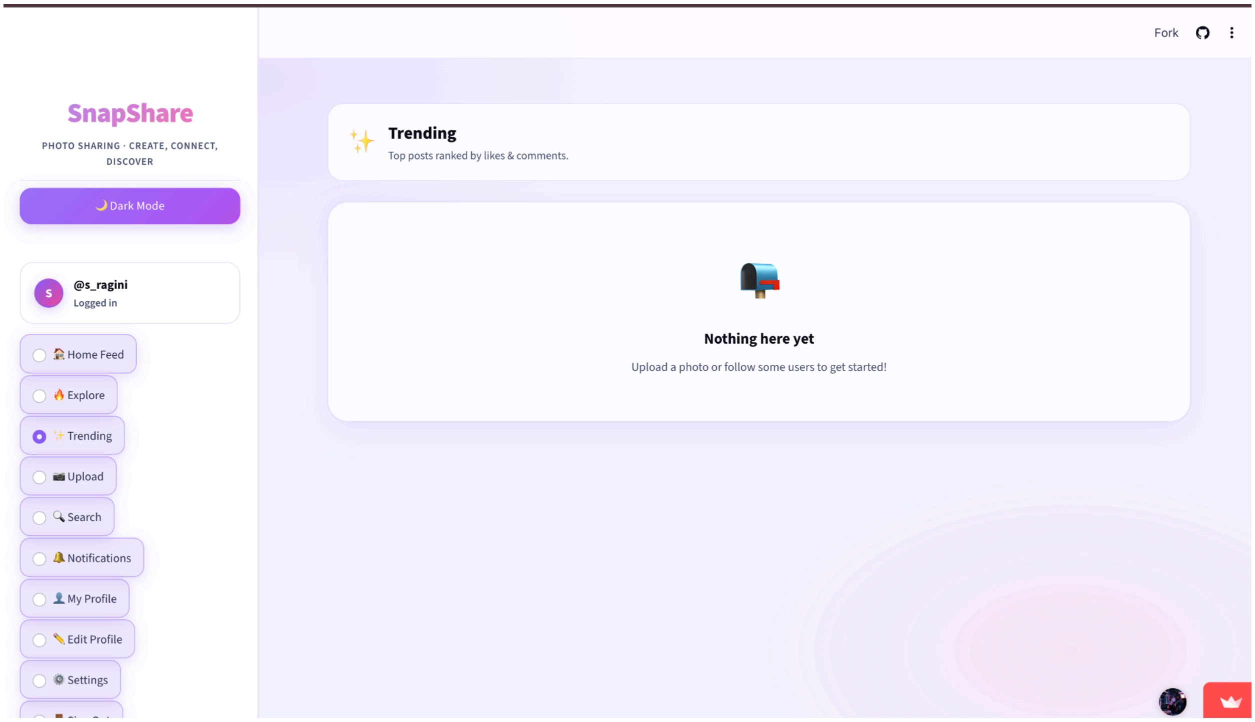


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
@s_ragini
Logged in
🏠 Home Feed
🔥 Explore
✨ Trending
📷 Upload
🔍 Search
🔔 Notifications
👤 My Profile
✏️ Edit Profile
⚙️ Settings
🚪 Sign Out
Fork
NETWORK & EXPLORE
Discover Creators
Search the community, follow trending designers, or connect with recommended profiles.
🔍 Search by username or bio snippet...
A Quiet Universe
It looks like you are the first pioneer here! To make exploration exciting, you can generate simulated creator accounts complete with bios and follow connections instantly.
⚡ Generate Demo Creators

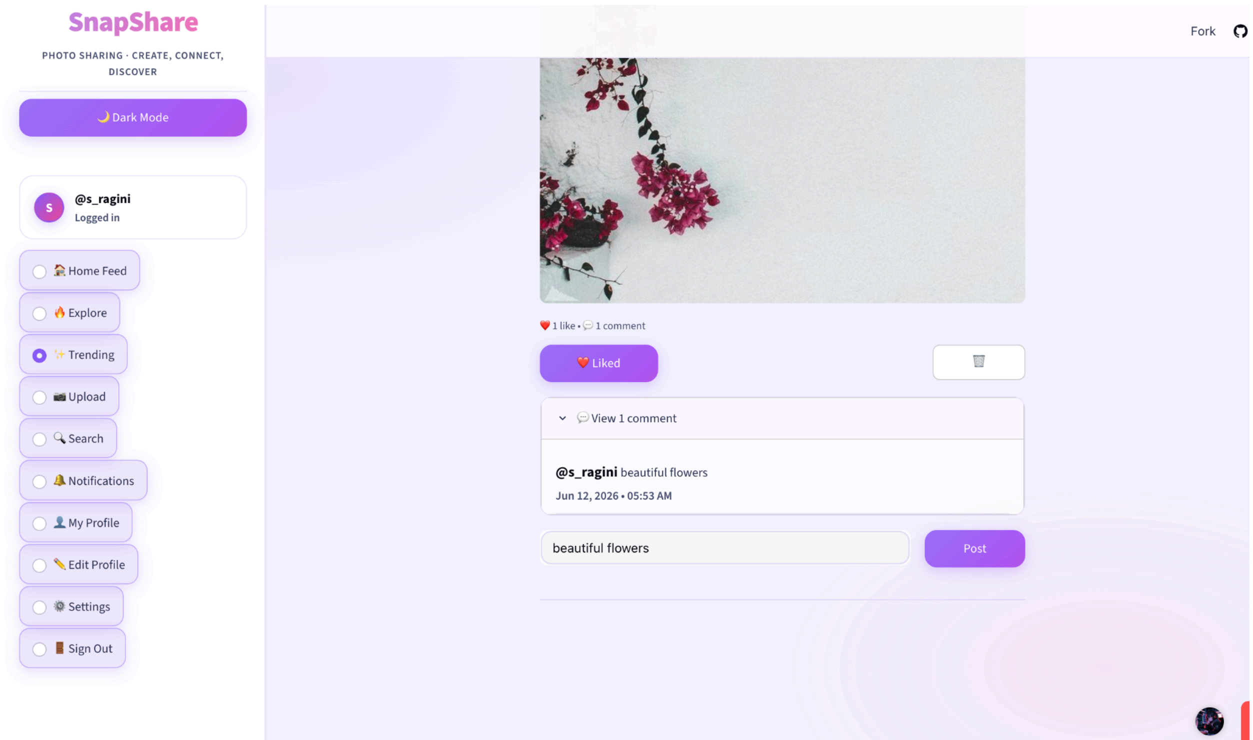
Setting page

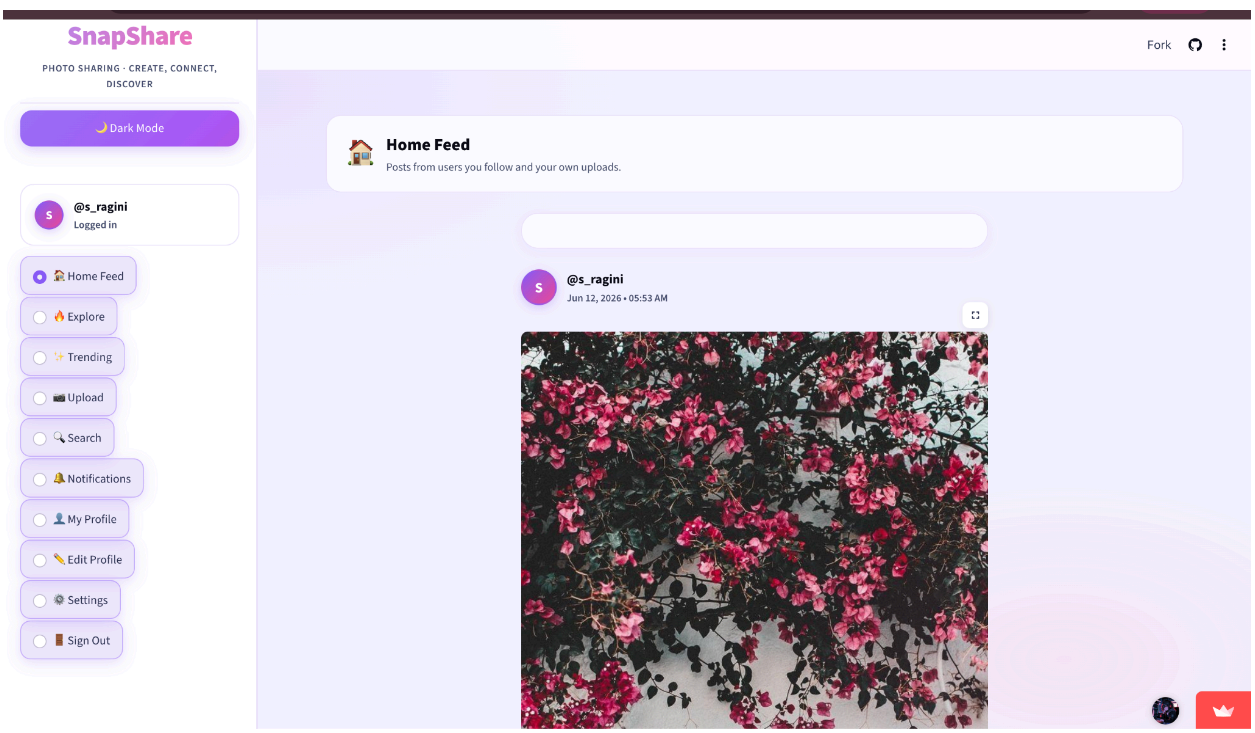


Trending page

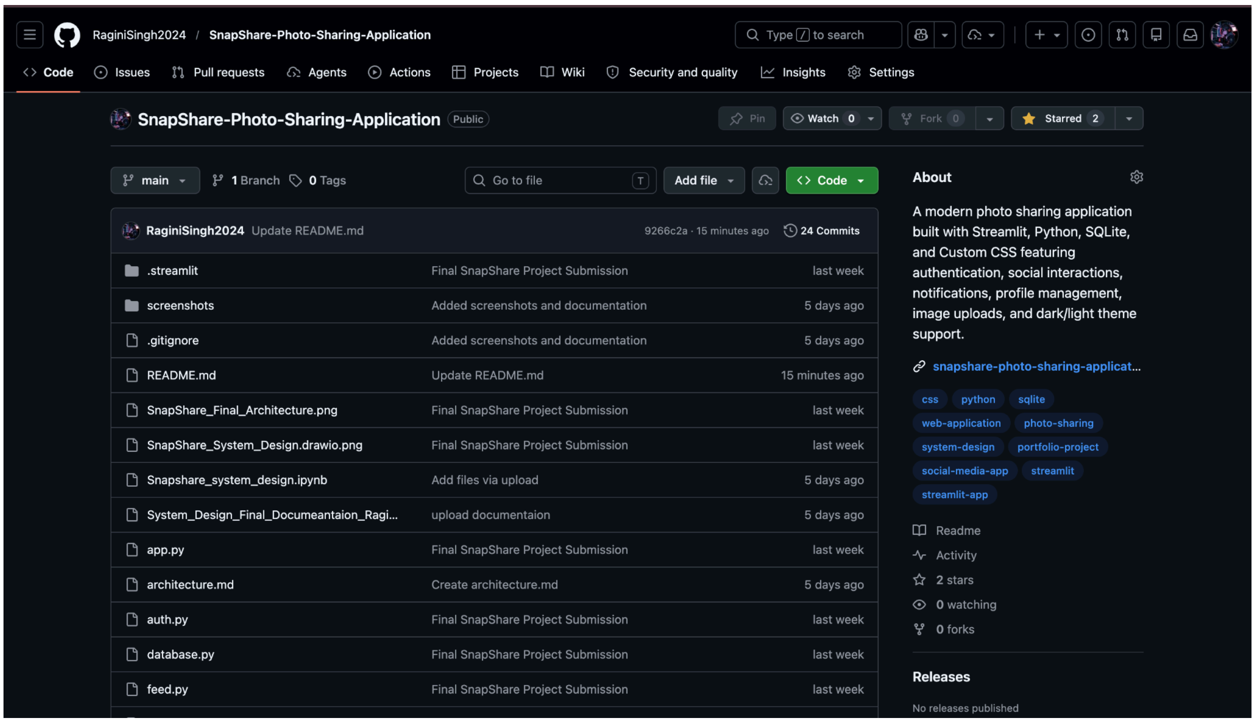


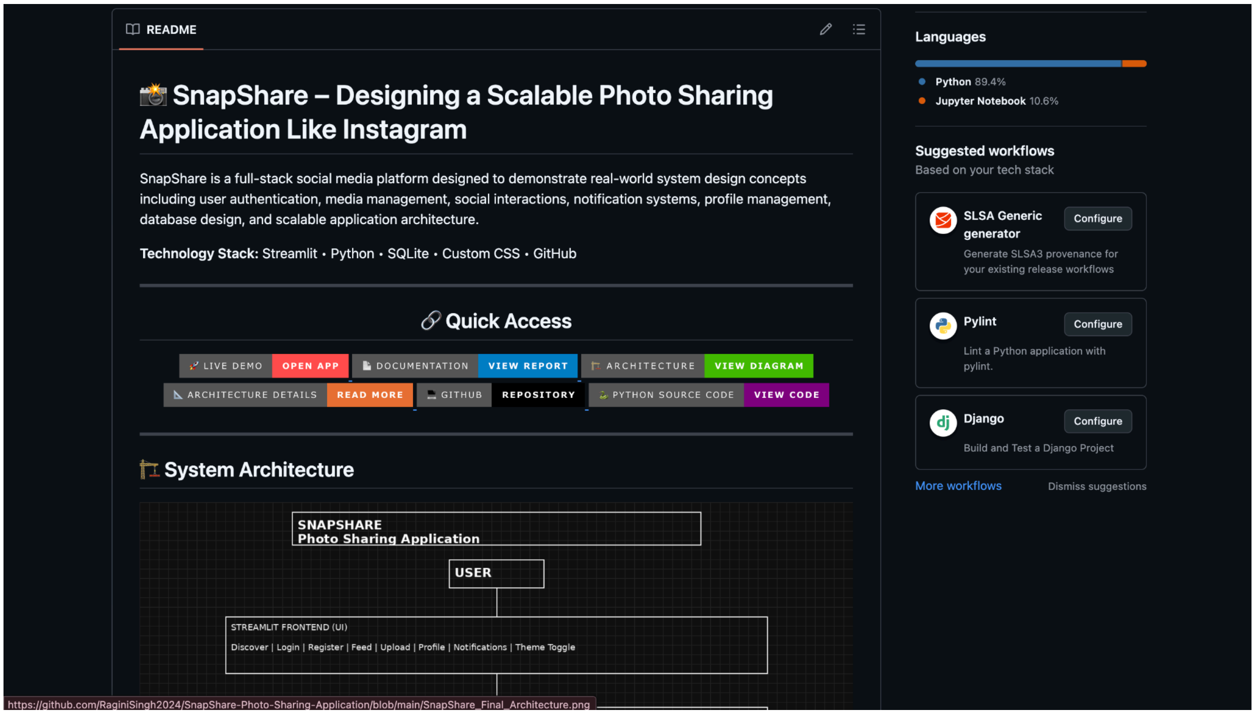
Upload page

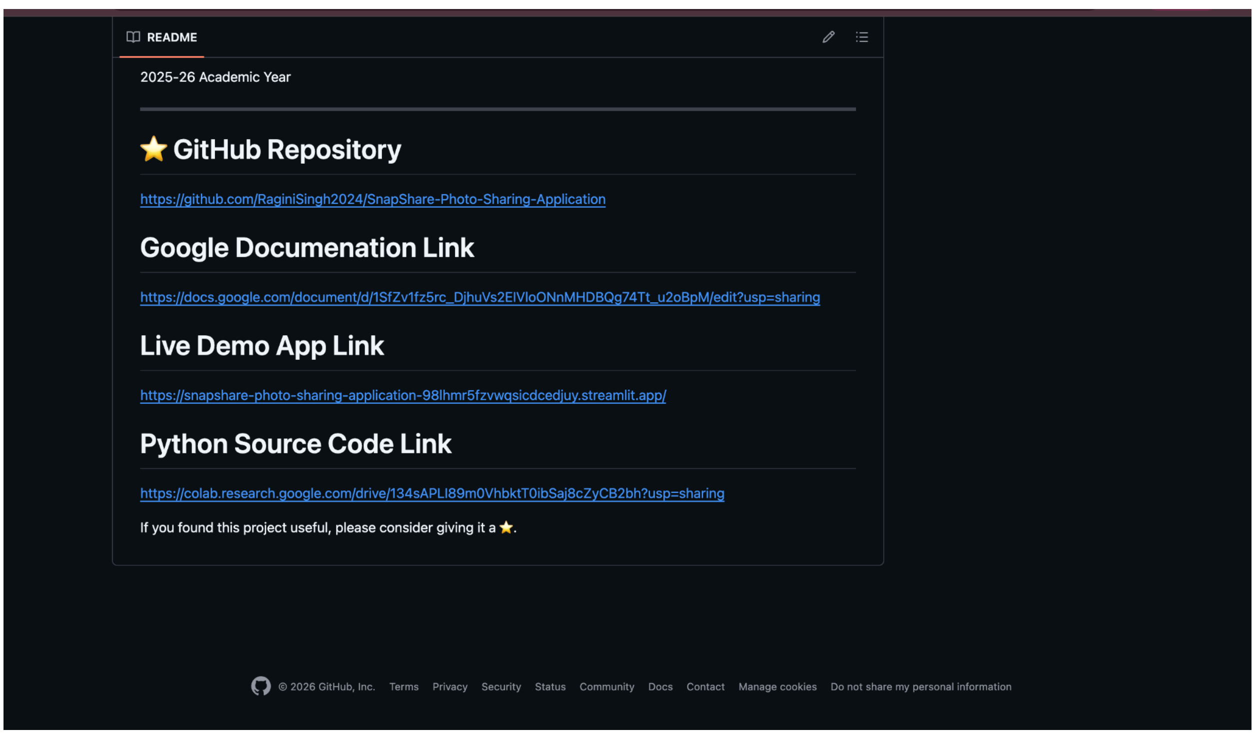

SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
@s_ragini
Logged in
🏠 Home Feed
🔥 Explore
✨ Trending
📹 Upload
🔍 Search
🔔 Notifications
👤 My Profile
✏️ Edit Profile
⚙️ Settings
🚪 Sign Out
Fork
❤️ 1 like • 💬 1 comment
❤️ Liked
💬 View 1 comment
@s_ragini beautiful flowers
Jun 12, 2026 • 05:53 AM
beautiful flowers
Post


SnapShare
PHOTO SHARING · CREATE, CONNECT, DISCOVER
🌙 Dark Mode
@s_ragini
Logged in
🏠 Home Feed
🔥 Explore
✨ Trending
📹 Upload
🔍 Search
🔔 Notifications
👤 My Profile
✏️ Edit Profile
⚙️ Settings
🚪 Sign Out
Fork
Home Feed
Posts from users you follow and your own uploads.
@s_ragini
Jun 12, 2026 • 05:53 AM

Github Repo SS







Scalability Considerations

For production-scale deployment, the following upgrades can be implemented:

- PostgreSQL Database
- Redis Cache
- AWS S3 Storage
- CloudFront CDN
- Load Balancer
- Containerization with Docker
- Kubernetes Orchestration
- Microservices Architecture

Security Features

- SHA-256 Password Hashing
- Session-Based Authentication
- Input Validation
- File Type Validation
- Image Upload Restrictions
- SQL Query Protection

🔮 Future Scope

- Real-Time Messaging
- AI-Based Recommendations
- Mobile Application
- Story Feature
- Reels Support
- Cloud Storage Integration
- Real-Time Notifications
- Analytics Dashboard

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Repository

If you found this project useful, please consider starting the repository.

GitHub Repository:

https://github.com/RaginiSingh2024/SnapShare-Photo-Sharing-Application

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THANKYOUU!!!!!